

Machine heuristic in algorithm aversion: Perceived creativity and effort of output created by or with artificial intelligence

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ABSTRACT

Products and services labeled as AI-created are perceived differently than identical ones attributed to humans, reflecting the phenomenon of algorithm aversion. This perception gap is partly explained by individuals' tendency to attribute lower effort and creativity to work produced by AI algorithms. Through two pre-registered experiments, we investigate how individuals evaluate creative content when it is labeled as created by AI, humans, or human-AI collaboration. While previous research has established algorithm aversion across various domains, our findings reveal nuanced patterns in creative contexts. Individuals consistently rate content labeled as AI-authored as less creative and less favorable compared to content labeled as human-authored. This effect is driven by a sequential mediation: perceived effort influences perceived creativity, which in turn shapes overall attitudes. Importantly, content labeled as human-AI collaboration performs significantly better than AI-only content, suggesting a potential mitigation strategy. Our main theoretical contribution is the identification of the machine heuristic (MH), individuals' preexisting beliefs about AI capabilities, as a key moderator. Individuals with stronger MH beliefs attribute more effort to AI-generated work, which positively influences perceptions of creativity along with behavioral intentions. Using Netflix show descriptions as stimuli, we show these effects extend beyond attitudes to impact behavioral outcomes like willingness to watch. These results expand algorithm aversion theory by establishing MH as an important boundary condition and highlighting human-AI collaboration as an effective mitigation approach, offering valuable insights for creative industries integrating AI while maintaining positive perceptions among users.

1. Introduction

Breakthroughs in generative artificial intelligence (AI) technologies have moved beyond mere technical advancements. From GPT language models to SORA's video generation capabilities (OpenAI, 2024), these systems now mark the increasing integration of AI into creative processes traditionally dominated by humans, creating new forms of human-AI interaction across diverse domains (Mei et al., 2025; Messer, 2024). While these technologies offer immense potential to enhance efficiency and innovation (McKinsey, 2023), they have simultaneously sparked public and scholarly skepticism regarding the value and perception of AI-generated work. This skepticism has become a key focus of psychological inquiry within human-AI interaction. It aligns with the established concept of algorithm aversion. This bias describes a tendency to prefer human decision-making or craftsmanship over

algorithmic alternatives, even when outputs are of identical quality (Dietvorst et al., 2015). This preference often persists even in contexts where algorithms are demonstrably outperforming humans (Dietvorst et al., 2015). This aversion may be particularly relevant in creative domains where unique human attributes are perceived as central.

Understanding human psychological responses and reception of AI-generated content is becoming increasingly important as AI continues to play a larger role in creative industries. This is exemplified by the resolution of the longest-ever Hollywood strike, which concluded only after establishing clear boundaries for AI in screenwriting: AI could serve as a writing assistant but not replace human writers (Anguiano, 2023). This negotiated settlement mirrored broader societal concerns about AI's expanding capabilities across multiple industries (Monteith et al., 2024). Regulatory frameworks now increasingly mandate transparency in AI-created content, with platforms like TikTok (TikTok,

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2023), YouTube (Duffy, 2023), and various governments (European Parliament, 2023; Luo & Dan, 2023) requiring clear labeling of AI involvement. This increasing demand for transparency reflects a societal awareness that the source of creative content significantly shapes human perception within evolving human-computer interactions. To fully grasp how individuals interpret these authorship signals and what cognitive mechanisms drive their reactions to AI-created or collaboratively produced content, we need more sophisticated psychological investigation.

While prior research has begun to map human responses to AI use in various domains, ranging from medical diagnoses (Longoni et al., 2019; Monteith et al., 2024) and product recommendations (Wien & Peluso, 2021) to artistic production (Magni et al., 2023; Messer, 2024), the full complexity of psychological reactions to AI's creative role remains poorly understood. This knowledge gap matters significantly because creativity has traditionally represented a uniquely human capability, associated with specific cognitive abilities like intuition, emotional intelligence, and intentionality that seemed beyond AI's capabilities (Huang & Rust, 2022). Now, advanced generative AI produces outputs remarkably similar to human creativity (Mei et al., 2025). Raffloer and Green (Raffloer & Green, 2025) found that audiences were equally or more immersed in stories written by AI than those written by humans, but only when the source of authorship was not made salient. These findings challenge established assumptions and create an urgent need to investigate how individuals perceive, judge, and respond to creative works attributed to humans, AI systems, or collaborative efforts. This research is particularly important for investigating how perceptions shift even when the underlying creative output remains the same (Hermann & Puntoni, 2024).

Current research on attitudes toward AI-created content typically compares purely human-made versus purely AI-generated work, and findings consistently show viewer skepticism toward content with AI labels (Hattori et al., 2024; Magni et al., 2023; Raffloer & Green, 2025). However, this binary comparison overlooks an increasingly common reality in human-AI interaction: the emergence of human-AI collaborative creation, where human creators use AI as tools rather than complete substitutes (Messer, 2024). This collaborative scenario represents an important yet underexplored middle ground in psychological research. It raises critical questions about how individuals perceive creativity when AI is presented as a collaborator rather than the sole creator, and whether they attribute different levels of genuine effort and innovation to identical creations based solely on the attributed source (human, AI, or human-AI collaboration).

Individual cognitive differences may influence these variable responses. One such difference is the machine heuristic (MH), a concept that describes the mental shortcuts, or rules of thumb, that people use when assessing machine behavior (Sundar & Kim, 2019). The heuristic is triggered when individuals perceive that they are interacting with a machine, leading them to automatically apply common stereotypes about machine capabilities. These stereotypes often include beliefs that machines are more objective, precise, and reliable than humans (Sundar & Kim, 2019). MH describes the predetermined beliefs individuals hold about machine capabilities (Yang & Sundar, 2024). For example, related research shows some individuals inherently view AI as highly accurate or objective, which influences their trust and credibility assessments (Henestrosa & Kimmmerle, 2025). While researchers have primarily studied such MH constructs concerning trust or accuracy in general human-computer interaction (Henestrosa & Kimmmerle, 2025), their influence on perceiving creative production remains less explored. Understanding how specific MH facets, particularly beliefs about AI's capacity for effort and creativity, influence evaluations of AI-generated creative works is increasingly important. This understanding can provide valuable insights into the psychological mechanisms underlying human acceptance or rejection of creativity attributed to AI or human-AI collaboration.

In the current paper, we examine perceptions of content labeled as AI-generated in a creative context. We conducted two pre-registered

experiments, specifically using identical Netflix show descriptions as stimuli across conditions, to investigate reactions when the content is labeled as originating from AI, a human, or a human-AI collaboration. This entertainment setting is particularly suitable as it reflects a domain where human consumption of generative AI is rapidly evolving and highly debated (Anguiano, 2023).

2. Theoretical background

2.1. Algorithm aversion

The use of AI in traditionally human domains, such as creative work, often encounters skepticism concerning the value and nature of AI-generated output. A core theoretical framework for understanding this human preference for non-AI alternatives is algorithm aversion (Mahmud et al., 2022). This bias is particularly pronounced in areas requiring subjective judgment and emotional intelligence, qualities often associated with human capabilities (Huang & Rust, 2022). Despite AI systems offering greater efficiency and lower costs (McKinsey, 2023), individuals remain skeptical of their ability to replicate human sensitivity and judgment in various processes (Longoni et al., 2019).

Algorithm aversion manifests across different contexts, as Table 1 illustrates. In practical utilitarian domains such as medical diagnosis, individuals resist AI recommendations even when presented with evidence of equal or superior accuracy (Longoni et al., 2019). Research by Van Esch and colleagues (Van Esch et al., 2019) revealed that job applicants resist AI-driven recruitment, though this resistance diminished when AI was positioned as assisting rather than replacing human decision-makers. Similarly, Wien and Peluso (Wien & Peluso, 2021) demonstrated that in hedonic contexts where emotional connection is crucial, the preference for human recommenders was particularly strong. Their research also revealed that adding human-like qualities to AI systems helped reduce this effect, suggesting that perceived humanness significantly influences acceptance. The complexity of these AI perceptions is further demonstrated by research on AI label effects. For instance, while an AI label could directly increase user engagement through a novelty effect (Chen et al., 2025), it also simultaneously created a negative indirect effect by lowering perceptions of content quality (Chen et al., 2025).

Research suggests that how AI processes are communicated shapes user acceptance. Chen and colleagues (Chen et al., 2024) found that explaining the calculative process of an algorithm can trigger feelings of 'meta-dehumanization', the perception of being treated as less than human, which increases algorithm aversion. This indicates that the way in which the process of an algorithm is presented is important. However, hybrid approaches combining human and AI capabilities may help bridge this acceptance gap by reframing the process as collaborative rather than purely algorithmic. Advisory contexts demonstrate that presenting AI outputs as collaborative efforts with substantial human involvement boosts individual receptivity (Zhang et al., 2024). Though, the effectiveness of such hybrid approaches, particularly human-AI collaboration in artistic content creation, remains less explored.

The intersection of AI and creativity thus presents a unique context for examining algorithm aversion. While creative domains involve qualities traditionally seen as human (Huang & Rust, 2022) and may amplify aversion effects compared to analytical tasks, the potential for mitigation through human-AI collaboration requires specific investigation. As generative AI technologies advance rapidly (OpenAI, 2024), understanding how individuals respond to varying levels of AI involvement in creative production becomes increasingly important.

As summarized in Table 1, previous research has established clear patterns of algorithm aversion across various domains. Our research extends this work by specifically examining human-AI collaboration in a creative context and investigating how psychological differences, specifically the machine heuristic, might moderate individual perceptions. To visually encapsulate the core relationships and mechanisms we

Table 1
Literature review table.

Source	Context	Relevant Findings
Dietvorst et al. (2015)	Weather Forecasts	Consumers evaluate human outputs higher than those from algorithms even if the algorithmic output is identical or outperforms the human.
Longoni et al. (2019)	Medical Service AI	Resistance to medical AI can be eliminated or mitigated when AI care supports rather than replaces decisions made by human providers.
Van Esch et al. (2019)	Recruitment	Potential job candidates resist to recruitment processes that rely on AI, but communicating Human-AI collaboration seems to mitigate the effect.
Chen & Sundar (2019)	Human-Computer interaction	Users with higher beliefs in machine heuristic are less likely to lose trust in AI after observing biases, viewing them as isolated incidents.
Judge et al. (2020)	Creativity and labor	Folk psychology and theories show that people value the mental and physical effort that goes into artifact creation.
Wien and Peluso (2021)	Recommendation	Human recommenders are preferred to AI in hedonic product categories because humans elicit stronger mentalizing responses. However, humanizing the AI mitigates this effect.
Nie and Huang (2023)	Service evaluation	Humanoid robots have a negative impact on service evaluation, where perceived effort is an explanatory mediating factor.
Magni et al. (2023)	Product Creativity	Consumers attribute lower effort and creativity to products made by AI compared to humans.
Tigre Moura et al. (2023)	Art Preference	Full automation or co-creation increased the perception of tangible products but not intangible ones. Full automation reduced the emotional value of soundtracks and artworks.
Choi et al. (2024)	Effort	People attribute less effort to AI systems, which leads to lower customer satisfaction.
Hattori et al. (2024)	Product Creativity	Consumers rate lower creativity to useful products generated by AI, than those labeled as fully human-made.
Zhang et al. (2024)	AI Advice	People perceive AI advice as more helpful when human input is highlighted, and the effect is mediated by increased perception of understanding.
Messer (2024)	AI Art Collaboration	Co-created art liked less, seen as less authentic, especially when AI implements. Reduced authenticity is key. Artists using AI admired less. Effort perception matters.
Mei et al. (2025)	AI Writing Creativity	Using AI enhances creative output performance but diminishes user experience (value, enjoyment) and reduces perceived effort from the author's side.
This Paper	Creativity	Algorithm aversion in individuals is moderated by machine heuristic and can be mitigated by labeling AI involved outputs as Human-AI collaborations.

propose to investigate, Fig. 1 presents our conceptual model. This model outlines how different authorship labels are hypothesized to influence individuals' perceptions and behavioral intentions, both directly and indirectly through perceived effort and creativity, and how these processes may be moderated by individual differences in machine heuristic beliefs. A detailed theoretical development of each component follows in Section 2.

2.2. Perceived creativity and attitudes

Creativity is the ability to create ideas or outcomes that are both novel and appropriate to fulfill goals (Amabile, 2011). Until recently, researchers and the public viewed creativity as a uniquely human trait that separates us from AI systems. The emergence of generative AI has dramatically altered this perspective. Modern AI now composes entirely new music (Agostinelli et al., 2023), creates original artwork (Betker et al., 2023), and produces realistic images and complex videos from text prompts (OpenAI, 2024). These systems also craft coherent stories using LLM-powered tools like ChatGPT (OpenAI, 2022), occasionally reaching quality levels comparable to skilled human work (Mei et al., 2025). Yet, whether individuals perceive such works as truly creative remains insufficiently explored. Since such perceptions crucially influence how individuals evaluate creative works, we need focused research on perceived creativity from the consumer perspective.

Research indicates that individuals assign lower creativity ratings to products labeled as AI-generated compared to identical products attributed to humans (Hattori et al., 2024; Magni et al., 2023; Raffloer & Green, 2025). This suggests that perceived creativity is not determined solely by the intrinsic qualities of the work but also on who supposedly created it. In an educational context, parents favor human tutors over AI tools specifically for creative tasks, linking this choice to a belief in the superior creative competence of humans (Shao et al., 2025). Tigre Moura and colleagues (Tigre Moura et al., 2023) found that full automation significantly reduced the emotional value consumers attributed to creative works like soundtracks and artworks. This devaluation, specific to creative products, highlights how automation uniquely affects perceptions in these domains, potentially stemming from lay theories about creativity (Magni et al., 2023). These lay theories suggest that individuals view creativity through distinct pathways: the 'insight pathway,' characterized by sudden, intuitive realizations often associated with human cognition (requiring qualities like intuition, emotional understanding, and intentionality), and the 'persistence pathway,' involving systematic effort sometimes associated with machine processing (Baas et al., 2015; Lucas & Nordgren, 2022). Though both approaches can produce creative results, most individuals prefer and value the insight pathway (Judge et al., 2020), believing that true creativity requires inherently human qualities connected to our distinctive cognitive structure, which fundamentally differs from current AI capabilities. This preference likely explains why individuals sometimes view AI-generated content as lacking authenticity (Messer, 2024) or humanness (Jia et al., 2024), contributing to lower creativity assessments.

Current research primarily compares purely human-created versus purely AI-generated work, neglecting how individuals perceive creative content from human-AI collaboration. Given the increasing prevalence of AI assisting human creators, it is important to explore whether combining the efforts of humans and AI influences perceptions of creativity. We aim to determine whether human-AI collaboration enhances perceived creativity compared to solely human efforts. Specifically, we want to explore if leveraging AI as an innovative tool actually improves creative outcomes, or whether collaborative output falls somewhere between purely human and purely AI creations, as we suspect. Extending previous findings to this collaborative context is crucial, making this investigation an important next step. This gap provides the novelty for our investigation, as we examine how individuals evaluate creativity in the context of streaming shows attributed to AI, human, and human-AI collaboration.

Based on how creativity perceptions fundamentally influence individual evaluations of creative work, and building on established patterns of algorithm aversion, we anticipate these effects will appear in both creativity assessments and overall attitudes. Therefore, we hypothesize.

H1. Shows written by humans will be perceived as more creative, followed by those written by a combination of AI and humans, and lastly those written by AI alone.

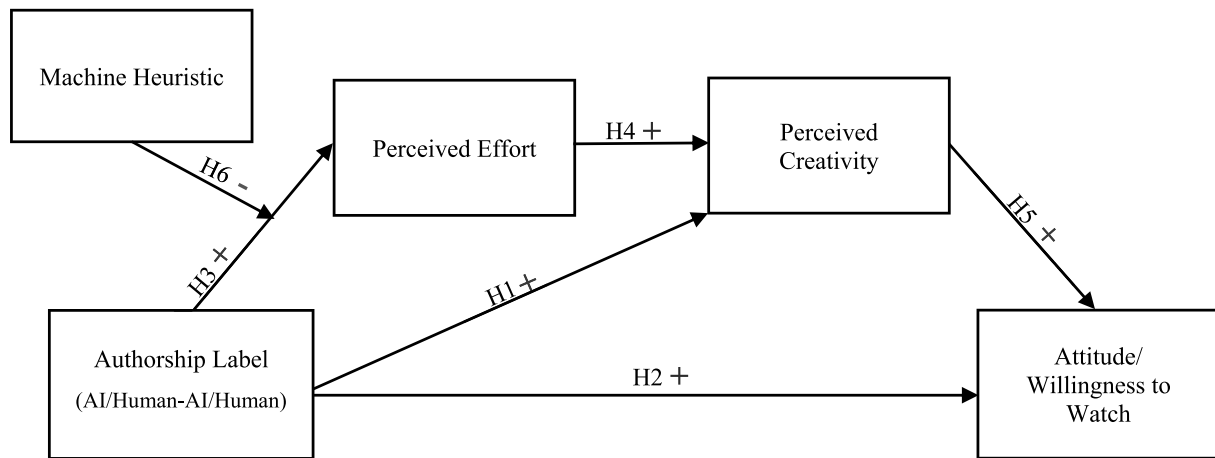


Fig. 1. Conceptual model

Note. (+) Indicates a hypothesized positive relationship; (−) indicates a hypothesized negative relationship. "Authorship Label" refers to the experimental conditions where content was labeled as authored by a human, AI, or human-AI Collaboration. H1, H2 and H3 represent hypothesized direct effects. H4, and H5 depict the hypothesized serial mediation pathway. H6 illustrates the hypothesized moderating effect of machine heuristic.

H2. Audiences will exhibit more favorable attitudes toward shows written by humans, followed by those written by a combination of AI and humans, and lastly those written by AI alone.

2.3. Effort

Drawing from the service literature, a key driver of individual preference for human services is the perception that AI systems and robots expend minimal or no effort (Nie & Huang, 2023). Effort, defined as the physical labor, ingenuity, or skill invested in the creation process (Judge et al., 2020), is deeply intertwined with lay theories of creativity. These theories suggest that valuable creative work requires deliberate effort and intentionality, qualities that individuals inherently associate with human rather than AI agents (Huang & Rust, 2022). The perceived absence of effort in AI may stem from folk beliefs about machines working automatically and efficiently, lacking the struggle and persistence that individuals associate with genuine creative achievement. Indeed, individuals using AI tools like ChatGPT for creative tasks report experiencing reduced effort themselves (Mei et al., 2025), which aligns with the viewers' potential perception of lower effort invested in the final AI-generated content.

Numerous studies consistently show that individuals attribute less effort to AI systems than to human counterparts (Lucas & Nordgren, 2022), 36]. This bias likely stems from perceptions that AI functions automatically or possesses fundamentally different capabilities than human cognition (Nie & Huang, 2023). Consequently, even when outputs are identical, AI-driven services or creations often receive lower evaluations partly due to this reduced perception of effort (Nie & Huang, 2023).

This effort bias extends beyond service evaluations and directly impacts perceptions of creativity in consumer behavior. Individuals consistently perceive AI as exerting less effort than humans when creating artifacts (Magni et al., 2023). This difference in perceived effort significantly influences creativity evaluations, with lower perceived effort leading to lower perceived creativity (Magni et al., 2023). However, while previous findings have established this effort bias in binary comparisons between AI and human creators, the impact of human-AI collaboration on effort perceptions remains largely unexplored, particularly in entertainment content creation. As streaming platforms increasingly experiment with AI in content development, understanding how individuals perceive the effort invested in collaborative human-AI creative processes becomes necessary.

Based on these findings, we hypothesize that.

H3. The Human and Human-AI authorship condition leads to higher perceived effort than the AI condition.

H4. Perceived effort mediates the authorship effect on perceived creativity.

H5. Perceived effort and perceived creativity mediate the effect of Human authorship compared to AI or collaboration on attitude, through a serial mediation.

2.4. Machine heuristic

Individual differences in pre-existing beliefs about AI may significantly influence how AI-generated or collaborative content is perceived. One key theoretical lens for understanding these variations is the machine heuristic. The MH construct, focusing on individuals' comparative assessment of machine versus human capabilities, has been validated as distinct from related concepts like technology dependence or automation bias (Hattori et al., 2024). While related, MH specifically captures the beliefs about inherent abilities, not just reliance or trust patterns. This focus on pre-existing beliefs aligns with other research showing that individual traits like 'technological optimism' moderate AI technology adoption (Yang et al., 2025).

The moderating role of such heuristic beliefs about AI has been demonstrated in human-computer interaction. For example, related research using a similar heuristic framing found that individuals who believe AI is inherently more accurate and less biased are less likely to lose trust when encountering flawed AI output, suggesting MH can act as a psychological buffer against negative information (Henestrosa & Kimmerle, 2025). Similarly, Chen and Sundar (Chen & Sundar, 2024) found that users with stronger general MH beliefs maintained trust despite encountering biases in AI.

While the work by Henestrosa and Kimmerle (Henestrosa & Kimmerle, 2025) focused on MH related to accuracy and its impact on credibility, our study investigates a different but related facet: MH concerning AI's capabilities for effort and creativity. MH becomes particularly relevant in creative contexts because it potentially moderates the effort-creativity evaluation pathway discussed previously. As previously assumed, individuals generally perceive AI as exerting less effort, which negatively impacts creativity ratings (Magni et al., 2023). We propose, however, that MH moderates this initial effort perception. Individuals with stronger MH beliefs (those who perceive AI systems as highly capable) may view AI's efficiency differently. Rather than interpreting it as a lack of effort that limits creativity, they may see it as a different and powerful capability (Hattori et al., 2024). For these

individuals, the perceived effort gap between human and AI creation might be smaller, consequently lessening the negative downstream impact on perceived creativity and attitudes. Conversely, individuals low in MH may adhere more strongly to the belief that true creative effort requires human-like struggle and persistence, thus perceiving a larger effort gap and evaluating AI-authored content more negatively.

Based on this, we hypothesize.

H6. The effect of Human authorship compared to AI or collaboration on perceived effort, is moderated by machine heuristic.

3. Empirical overview

We administered two experimental studies to test our hypotheses. To ensure the robustness of our findings and to rule out alternative explanations we employed two different experimental designs across our studies. Study 1 used a within-subjects design to assess changes in perceptions within the same participants. Recognizing that repeated measures can introduce demand effects, the second experiment employed a between-subjects design. This approach verified that the effects observed in the first study represented actual differences in individuals' perceptions while allowing deeper investigation of the underlying explanatory mechanisms.

3.1. Study 1

3.1.1. Methods

We conducted a within-subjects experiment to examine the effects of communicating three different authorship label types on perceived creativity and attitude. Entertainment show evaluation represents an ideal context for studying these effects, as viewers regularly engage in assessing and selecting content based on brief descriptions on platforms like Netflix. This context allows us to examine how different authorship labels affect attitudes in a familiar setting. Each participant was exposed to all three authorship conditions: AI, human-AI collaboration, and human. These authorship labels were presented in a random order across three different fictitious shows with three different genres: *Ecliptic Frontiers* (sci-fi), *Pulse of the Planet* (documentary), and *Night City Neighbors* (comedy). The experiment was designed and randomized using Qualtrics. The methods and analysis were pre-registered.

This within-subjects approach allowed us to present multiple stimuli while randomizing the authorship condition for each participant. By having everyone evaluate content under all authorship conditions across different genres, we prevented individual preferences from systematically skewing creativity and attitude assessments.

3.1.2. Procedure and participants

We commissioned a standard sample of 250 US-based respondents with English as their first language from Prolific's online panel, which resulted in 251 completed participations. In line with Prolific's fair payment standards, participants were compensated £0.50. The fully anonymized dataset is available as supplementary online material. The analysis was conducted on a final sample of 248 respondents, after excluding three for failing attention checks. The sample consisted of 50.8 % men and 47.6 % women, with the remaining participants identifying as non-binary, third gender or preferred not to say. The ages of the participants ranged from 19 to 70 years old.

A post-hoc power analysis using G*Power 3.1 indicated that the final sample size provided sufficient statistical power for the main within-subjects effect of authorship label. For our dependent variable, attitudes, the observed effect size (partial $\eta^2 = 0.120$, Cohen's $f \approx 0.369$) yielded an achieved power exceeding 0.999 ($\alpha = .05$, three measurements, assumed correlation = 0.5).

The study began with participants reading instructions for rating upcoming Netflix shows. Each participant viewed a photo and description of a fictitious show, as illustrated in Fig. 2. While the content of any given show's description remained consistent, we manipulated the authorship attribution, labeling it as created by either an AI system, a human, or a human-AI collaboration. After evaluating each show on perceived creativity and attitudes, participants repeated this process with two additional fictitious shows from different genres. This procedure ensured each participant encountered all three authorship conditions exactly once. We randomized which authorship label was assigned to each show and the sequence in which participants viewed these label-show combinations.

Perceived creativity was measured using a scale adapted from Magni et al. (Magni et al., 2023): 1-very uncreative to 7-very creative, 1-very unoriginal to 7-very original, and 1-very poor quality to 7-very good quality ($\alpha = .89$). Attitudes were measured with three items from MacKenzie and colleagues (MacKenzie & Lutz, 1989): 1-bad to 7-good, 1-unpleasant to 7-pleasant, and 1-unfavorable to 7-favorable ($\alpha = .96$). At the end of the study, participants provided demographic information about their age, gender, education, and income, a self-assessment of their perceived knowledge about AI (1-not knowledgeable at all to 7-extremely knowledgeable), and completed an attention check.

3.1.3. Results

To test the hypothesized effects of author type labels on show perception, two repeated measures ANOVAs were conducted with attitude and perceived creativity as dependent variables, corresponding to H1 and H2, respectively. Fig. 3's repeated measures ANOVA results

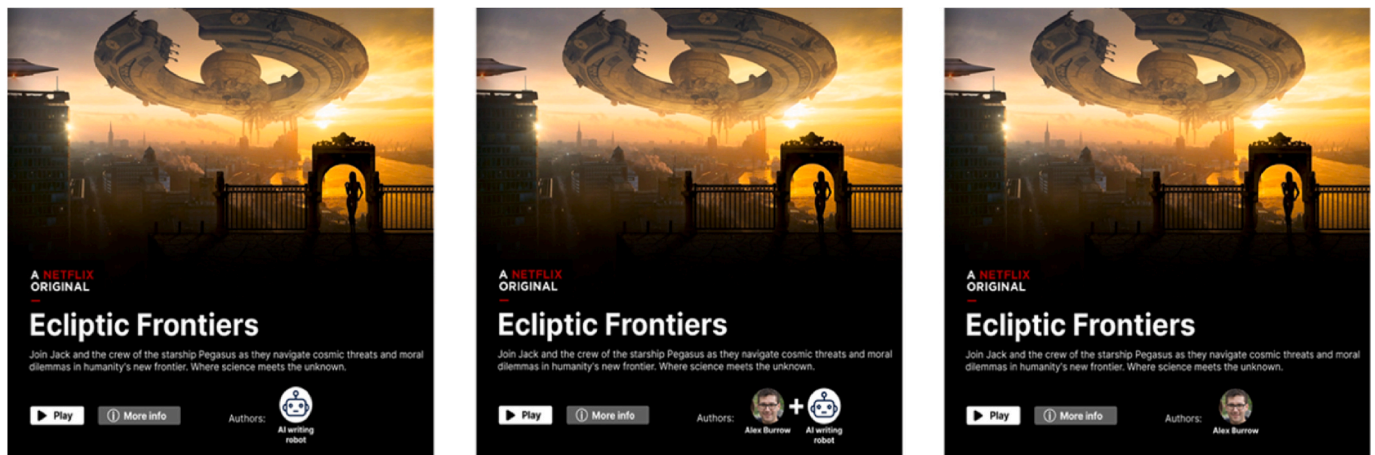


Fig. 2. The First Manipulation Stimuli with Differing Author Type Labels

Note. The two additional fictional shows being manipulated are detailed in the supplementary materials.

Perceived Creativity Scores for Different Author Types. Repeated measures ANOVA.

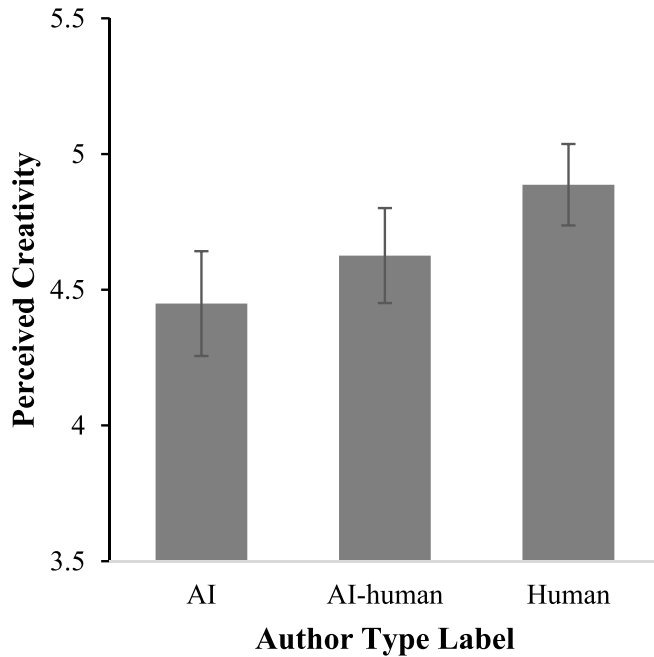


Fig. 3. Perceived Creativity Scores for Different Author Types. Repeated measures ANOVA.

Note. Error bars indicate 95 % confidence interval.

align with H1, illustrating that the human label significantly enhances perceived creativity of the show. The perceived creativity toward AI-only written shows ($M_{AI} = 4.45$), human-AI written shows ($M_{Human-AI} = 4.63$), and human-only written shows ($M_{Human} = 4.89$) displayed a progressive increase, with each step showing statistical significance (AI vs. human-AI: $\Delta = 0.18$, $p = .03$; human-AI vs. human: $\Delta = 0.26$, $p = .003$; AI vs. human $\Delta = 0.44$, $p < .001$). As depicted in Fig. 4, the analysis supports H2 with AI scoring the lowest on attitude toward the show ($M_{AI} = 4.43$) followed by human-AI ($M_{Human-AI} = 4.72$) and human scoring the highest ($M_{Human} = 5.20$). This indicates more positive attitudes with the human contribution. The mean differences between the author types were all significant at the 0.05 level (AI vs. human-AI: $\Delta = 0.29$, $p < .001$; human-AI vs. human: $\Delta = 0.48$, $p < .001$).

To account for potential genre effects, participants viewed shows from three different genres. Statistical analysis revealed that show genre did not significantly influence participants' perceived creativity or attitudes toward the show descriptions ($p = .823$), suggesting that our findings were consistent across different content types.

3.1.4. Discussion of study 1

The results of Study 1 reveals that attributed authorship significantly influences both perceived creativity and attitudes toward creative content, even when the underlying content remains identical. Shows labeled as human-authored were consistently rated as more creative and received more favorable attitudes compared to those labeled as originating from AI or human-AI collaboration. This finding extends previous research on algorithm aversion by showing that the effect persists in the context of narrative entertainment content, possibly reflecting underlying perceptions about the fundamental differences in human versus AI cognition (Monteith et al., 2024) or perceived humanness (Jia et al., 2024).

Importantly, our findings reveal a consistent pattern where Human-AI collaboration occupies a middle ground. While collaborative

Attitude Scores for Different Author Types. Repeated measures ANOVA.

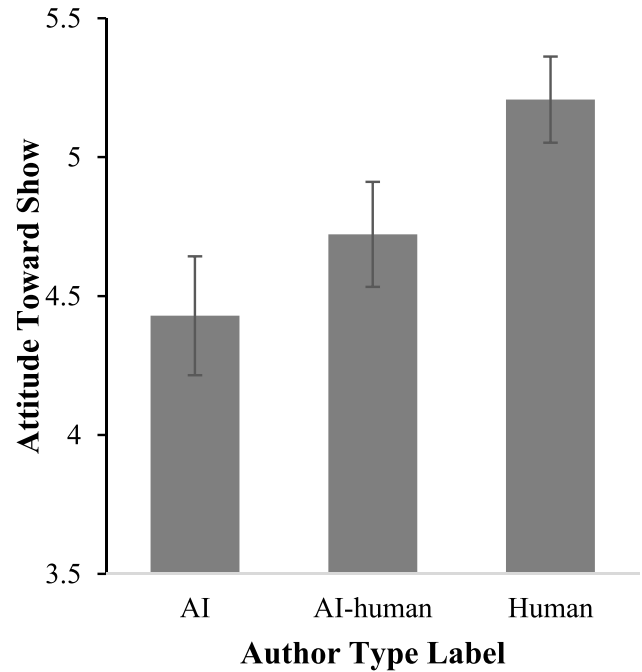


Fig. 4. Attitude Scores for Different Author Types. Repeated measures ANOVA.

authorship did not match the positive evaluations of purely human-authored content, it significantly outperformed the AI-only condition. This suggests that explicitly communicating human involvement in AI-assisted creative processes, a form of human-AI interaction, can partially mitigate algorithm aversion effects, a nuance also explored in contexts like co-created art (Messer, 2024).

The robustness of these effects across different show genres (sci-fi, documentary, and comedy) indicates that the influence of authorship labels on participant evaluations transcends specific content categories. This generalizability strengthens the evidence for a fundamental bias in how individuals evaluate AI versus human creative output when the source is made salient.

While Study 1 establishes the presence and pattern of these effects based on authorship labels, it leaves open the question of why individuals respond differently. Study 2 investigates the underlying mechanisms, particularly examining how perceived effort and individual differences in machine heuristic might explain these systematic variations in participant response.

3.2. Study 2

We conducted a one-factor between-subjects experiment to explore the underlying mechanisms behind the findings in Study 1. This study investigates the mediating role of creativity, perceived effort (Magni et al., 2023), and explores the moderating factor machine heuristic. Participants were randomly assigned to one of the three experimental conditions: AI-only, human, or human-AI. The experiment was designed and randomized using Qualtrics.

3.2.1. Procedure and participants

An a priori power analysis (G*Power 3.1) indicated that 489 participants were needed to achieve 90 % power to detect an anticipated small-to-medium effect size (Cohen's $f = 0.15$, $\alpha = .05$). While prior work by Magni et al. (2023) explored related biases concerning producer

identity, readily transferable effect size estimates for direct comparisons of human vs. AI on creativity in their most comparable between-subjects studies were not explicitly reported. Given our study's novel stimuli and investigation of the machine heuristic as a moderator, $f = 0.15$ was chosen as a suitable target for detecting meaningful effects for which prior precise estimates are limited.

The recruitment strategy for Study 2 mirrored that of Study 1, utilizing Prolific's online panel to source a US-based sample of native English speakers. In line with Prolific's fair payment standards, participants were compensated £0.50. The fully anonymized dataset is available as supplementary online material. A key difference was the implementation of an exclusion list: Prolific IDs from participants who completed study 1 were blocked to prevent re-participation and ensure sample independence. The final sample for analysis comprised 527 respondents ($M_{age} = 46.3$; 50.7 % females). A lower cutoff of 120 s ensured participants had sufficient time to process the stimuli and provide thoughtful responses. An upper cutoff of 10 min was applied to exclude those who may have been distracted or multitasking. The decision to implement time-based exclusions was in line with the pre-registered plan to ensure data quality. The average time spent on the intervention phase viewing the show description stimulus page was $M = 21.77$ s ($SD = 26.47$, $Min = 1.17$, $Max = 363.66$).

Similar to the first study, participants first read instructions about rating upcoming Netflix shows. The author manipulation was updated to include a visual representation of the writing process inspired by Zhang and colleagues (Zhang et al., 2024), in order to improve the readability and clarity of the stimuli without altering its purpose. Participants were then presented with a photo and a text description of a fictitious show that were written by either an AI author, human author, or a human-AI collaboration as authors (Fig. 5). The perceived creativity scale was adjusted to reflect the participants' subjective perceptions rather than objective ratings, as done by Clegg and colleagues (Clegg et al., 2024). Participants rated their agreement with four statements on a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree), including: "I feel the content of this show is creative," "The creativity of this show is high," "The story of this show seems creative," and "I see this show as creative" ($\alpha = .98$). Perceived effort was measured using an adapted scale from Magni and colleagues (Magni et al., 2023).

The belief in machine heuristic was measured using a scale adapted from Yang and Sundar (Hattori et al., 2024). Participants rated their agreement with four statements on a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree), including: "I trust machines more than

humans in performing scriptwriting," "Machines are more reliable than humans in writing and creating shows," "I believe machines are more competent than humans in writing and creating shows," and "Machines are better than humans in performing writing and creating shows" ($\alpha = .95$).

3.2.2. Results

The direct effect of the authorship label condition as hypothesized in H1 and H2 was confirmed for the human vs. AI and the human vs human-AI comparisons, with shows written by humans being perceived as more creative (human vs. AI: $\Delta = 0.38$, $p < .05$; human vs. human-AI: $\Delta = 0.40$, $p < .05$) and eliciting more favorable attitudes (human vs. AI: $\Delta = 0.44$, $p < .05$; human vs. human-AI: $\Delta = 0.38$, $p < .05$) than those authored solely or partly by AI. This effect replicated across both Study 1 and Study 2, providing robust support for the hypothesized difference between human-authored and AI-authored content.

To test H3, we conducted a univariate ANOVA with author type as the independent variable and effort as the dependent variable. Perceived effort of AI-only written shows ($M_{AI} = 3.64$), human-AI written shows ($M_{Human-AI} = 4.04$), and human-only written shows ($M_{Human} = 5.02$) showed a progressive increase, with each step showing statistical significance (AI vs. human-AI: $\Delta = 0.41$, $p < .013$; human-AI vs. human: $\Delta = 0.97$, $p < .001$; AI vs human: $\Delta = 1.38$, $p < .001$). These results provide support for H3.

To test H4, we conducted mediation analyses using the PROCESS module by Hayes (2018) with the authorship condition as the independent variable, perceived effort as the mediator, and creativity as the dependent variable. The mediation effect of perceived effort was significant for all comparisons as hypothesized (see Table 2).

Table 2

Mediation analyses for Authorship → Effort → Creativity.

Author comparison	Direct Effect (SE)	<i>p</i>	Indirect Effect (SE)	95 % CI
AI vs Human	−0.86 (0.13)	<0.0001	1.24 (0.16)	[0.94, 1.56]
AI vs Human-AI	−0.38 (0.12)	<0.01	0.36 (0.15)	[0.06, 0.66]
Human-AI vs Human	−0.49 (0.13)	<0.0001	0.89 (0.15)	[0.59, 1.19]

Note.

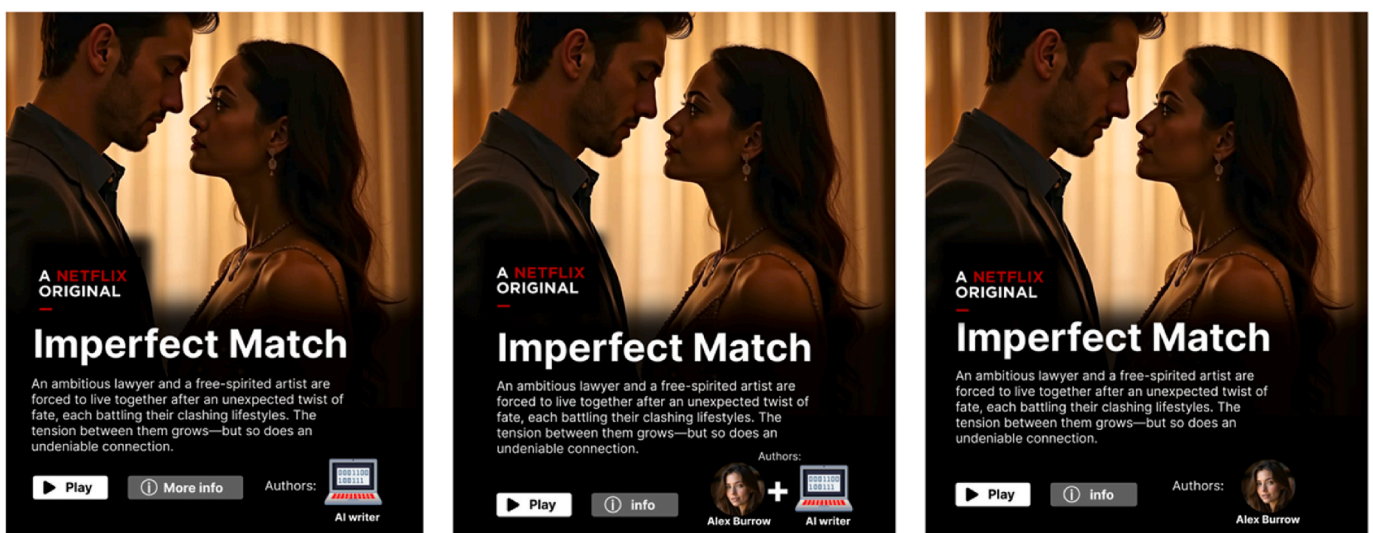


Fig. 5. The Author Manipulation with its Three Different Author Type Labels

Note. The image used for the show description was generated by using Flux.1 AI. Instructions are included in table S5 in the supplementary materials.

Serial mediation analyses using PROCESS model 6 (Hayes, 2018) supported H5, confirming that perceived effort and creativity sequentially mediate the relationship between authorship type and attitude. As shown in Table 3, the hypothesized indirect path through both mediators (Effort → Creativity → Attitude) was significant across all authorship comparisons ($b = 0.83$ to 0.25 , $p < .05$). While the total indirect effect was non-significant for the AI vs. human-AI comparison ($b = 0.07$, 95 % CI $[-0.26, 0.40]$), this resulted from opposing indirect effects: the serial mediation path remained significant but was counteracted by other indirect paths working in opposite directions (see Table 4).

To test H6, we conducted a moderation analysis using PROCESS model 1 (Hayes, 2018) to test whether the relationship between authorship type (AI, human-AI, human) and effort is moderated by the machine heuristic. The analysis revealed a significant interaction between authorship type and machine heuristic, $b = -0.17$, $p < .01$, 95 % CI $[-0.2827, -0.0625]$. This indicates that the effect of authorship type on perceived effort depends on the level of the machine heuristic, supporting our hypothesis. At low levels of the machine heuristic (MH = 1.00), the effect of authorship type on perceived effort was stronger ($b = 0.90$, $p < .001$). As MH increases, the effect size decreases but remains significant up to a score of MH = 4.55. Beyond this point, the effect becomes non-significant, as indicated by the Johnson-Neyman significance region, which shows that the relationship between authorship type and perceived effort is significant for 92.2 % of the sample.

Combining all results and hypotheses, we analyzed the moderated serial mediation model using PROCESS Model 83 to examine the impact of authorship type label (AI, human-AI, human) on attitudes through perceived effort and creativity, moderated by MH. The serial mediation pathway (Authorship type → Effort → Creativity → Attitudes) was significant and moderated by MH. At low MH, the pathway had a strong effect ($B = 0.55$, 95 % CI $[0.41, 0.70]$), diminishing at higher levels of MH ($B = 0.23$, 95 % CI $[0.12, 0.35]$). The index of moderated serial mediation ($B = -0.10$, 95 % CI $[-0.16, -0.05]$) confirmed that stronger MH beliefs reduce the influence of authorship type on attitudes via perceived effort and creativity. These findings suggest that MH beliefs significantly attenuate the perceived advantages of human authorship in terms of both effort and creativity, ultimately weakening its impact on attitudes.

We also conducted an identical serial mediation analysis, only swapping willingness to watch as the dependent variable, which represents a more downstream behavioral outcome compared to attitudes (See Fig. 6). The overall pattern of results remained consistent with the previous analysis, but some effect sizes differed slightly. The serial mediation pathway (Authorship type → Effort → Creativity → Willingness to Watch) was significant and moderated by MH. At low MH, the pathway had a strong effect ($B = 0.67$, 95 % CI $[0.50, 0.86]$), diminishing at higher levels of MH ($B = 0.29$, 95 % CI $[0.14, 0.43]$). The index of moderated serial mediation ($B = -0.13$, 95 % CI $[-0.20, -0.06]$) confirmed that MH beliefs attenuate the influence of human authorship on willingness to watch via effort and creativity. These findings indicate that MH not only moderates the impact of authorship on attitudes but also its downstream effects on behavioral intentions, underscoring the

Table 3
Serial mediation analyses for Authorship → Effort → Creativity → Attitude.

Author comparison	Direct Effect (p)	Total Indirect Effect (95 % CI)	Ind3 serial mediation (95 % CI)
AI vs Human	-0.19 ($p = .06$)	0.63 $[0.29, 0.97]$	0.83 $[0.62, 1.07]$
AI vs Human-AI	-0.02 ($p = .87$)	0.07 $[-0.26, 0.40]$	0.25 $[0.05, 0.46]$
Human-AI vs Human	-0.11 ($p = .09$)	0.49 $[0.18, 0.80]$	0.59 $[0.39, 0.81]$

Note. Table including other indirect paths is included in the supplementary materials.

Table 4
Hypotheses with results.

Study 1		Pre-registered	Result
H1	Shows written by humans will be perceived as more creative, followed by those written by a combination of AI and humans, and lastly those written by AI alone.	Yes	Supported
H2	Audiences will exhibit more favorable attitudes toward shows written by humans, followed by those written by a combination of AI and humans, and lastly those written by AI alone.	Yes	Supported
Study 2			
H3	The Human and Human-AI authorship condition leads to higher perceived effort than the AI condition.	Yes	Supported
H4	Perceived effort mediates the authorship effect on perceived creativity.	Yes	Supported
H5	Perceived effort and perceived creativity mediate the effect of Human authorship compared to AI or collaboration on attitude, through a serial mediation.	Yes	Supported
H6	The effect of Human authorship compared to AI or collaboration on perceived effort, is moderated by Machine heuristic.	Yes	Supported

Note. Pre-registration link for study 1 and study 2. H5 and H6 were registered as exploratory.

broad relevance of MH beliefs in shaping individuals' responses to AI-versus human-authored content.

3.2.3. Discussion study 2

The consistent finding of progressively higher perceived effort from AI-only, to human-AI collaboration, to purely human-authored content provides important insights into how participants evaluate AI-involved creative work. While Study 1 revealed a clear stepped pattern for direct effects on attitudes and perceived creativity, Study 2, while also showing this trend for effort evaluations, presented a more nuanced picture for direct effects on other outcomes when exploring underlying mechanisms.

Study 2's primary contribution lies in identifying these explanatory mechanisms. The serial mediation analysis confirmed that perceived effort is a crucial mediator of creativity evaluations, extending previous findings by Magni et al. (Magni et al., 2023). Participants rated outputs as more creative when they perceived higher levels of effort invested in their creation. This aligns with research showing that individuals using AI tools for creative tasks report experiencing less effort themselves (Mei et al., 2025), which likely translates to viewers perceiving lower effort in the final AI-generated output. Furthermore, the perceived effort involved can influence broader evaluations, such as the perceived authenticity of a creative piece (Messer, 2024). Our findings thus underscore that individual perception of the effort invested during the creation process heavily influences subsequent evaluations of creative work produced through human-AI interaction. This effort-creativity pathway aligns with other psychological mechanisms identified in the literature, such as meta-dehumanization, which can increase algorithm aversion (Zhang et al., 2024). Both mechanisms suggest that perceptions of how AI operates, whether as effortless or dehumanizing, can negatively impact user acceptance.

The moderating effect of machine heuristic on effort perceptions represents a significant contribution to algorithm aversion research and the study of human-AI interaction. While previous research has identified factors like general attitudes toward AI or technology familiarity as influencing algorithm aversion (Mahmud et al., 2022), our conceptualization of MH specifically captures individuals' beliefs about AI systems' inherent capabilities for tasks like creative production. This differs from, yet complements, other operationalizations of MH that focus on perceived accuracy or objectivity in different contexts (Henestrosa &

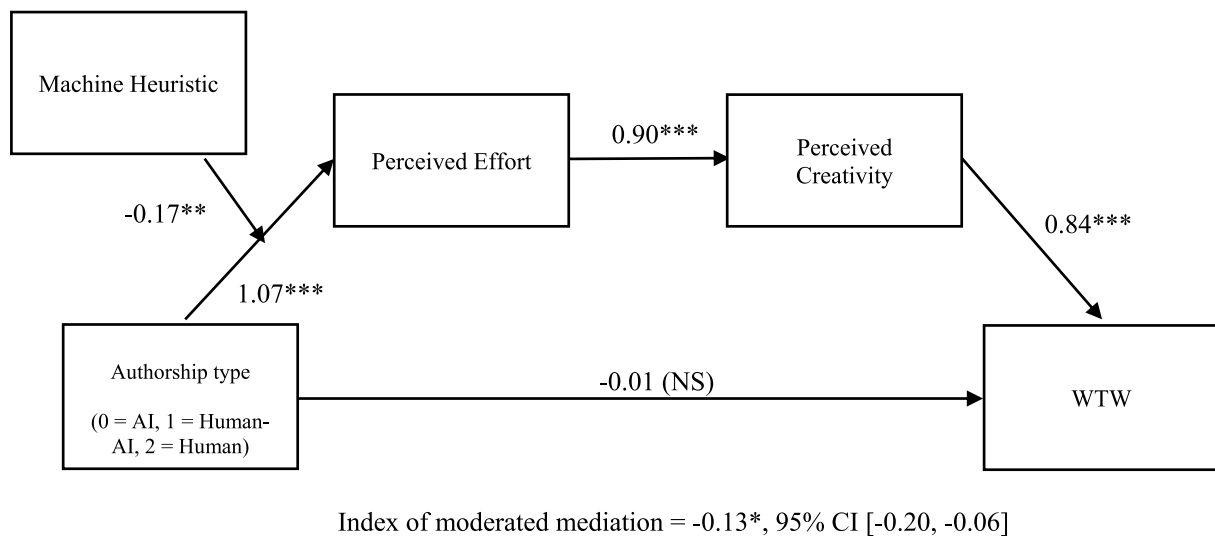


Fig. 6. Moderated Serial Mediation Analysis Using PROCESS Model 83

Note. $*$ = $p < .05$, $**$ = $p < .01$, $***$ = $p < .001$.

Kimmerle, 2025). Our analysis revealed that these capability-related beliefs uniquely moderate how attributed authorship affects perceived effort: as MH increased, the perceived effort difference between human-authored and AI-involved content diminished. For participants with low MH beliefs, the traditional hierarchy of perceived effort (human > human-AI > AI) was pronounced. However, for those high in MH, this difference attenuated significantly. This suggests that individual cognitive frameworks regarding the fundamental capabilities of AI systems substantially alter the psychological processing of effort when evaluating creative work attributed to different sources. This finding converges with Chen and colleagues (Chen et al., 2025), who found that users with high scores in 'perceived AI capabilities' were less likely to devalue AI-generated video content, providing robust evidence that individual cognitive frameworks about AI are a critical boundary condition for algorithm aversion.

The combined moderated serial mediation analysis reveals the full complexity of how machine heuristic shapes individuals' responses to AI-involved authorship. The results demonstrate that MH moderates not just the initial perception of effort, but the entire psychological pathway through which attributed authorship influences attitudes and behavioral intentions via perceived effort and creativity. For participants with low MH beliefs, the source label (human vs. human-AI vs. AI) strongly influenced their evaluations through this mediated chain. In contrast, participants with high MH beliefs exhibited much weaker effects of authorship labels on both attitudes and willingness to watch. This finding is particularly meaningful as it demonstrates that MH's influence extends beyond immediate perceptions to downstream behavioral outcomes relevant to human interaction with AI-generated or co-created media. The notable impact on willingness to watch suggests that these underlying cognitive heuristics about AI capability may be especially influential when individuals make actual consumption decisions.

4. General discussion

This study examines how individuals evaluate creative work produced by artificial intelligence, focusing on how the machine heuristic influences perceived effort, creativity assessments, and algorithm aversion toward AI-generated content. Creativity has long been considered a defining characteristic of human cognition, representing our ability to generate novel and valuable ideas that transcend routine thinking. Sophisticated generative AI systems have challenged this traditional understanding by demonstrating capabilities that increasingly blur the line between human and machine creativity. AI systems now compose music,

generate artwork (Messer, 2024), and write narratives that can match human-created work in technical quality (Mei et al., 2025). However, our research reveals that technical capability alone does not determine how humans perceive creative work. Rather, the attributed source of creativity fundamentally shapes how individuals evaluate and psychologically respond to these outputs, a core concern in human-AI interaction.

Through two pre-registered experiments examining participants' responses to streaming content descriptions, we found that explicitly labeling AI involvement in creative works consistently influences perceptions, even when the underlying content remains identical. This finding extends beyond simple preference effects, revealing a complex interplay between attributed authorship, perceived effort, and creativity evaluations. Our results consistently showed human authorship outperforming other conditions in enhancing perceived effort, creativity, and attitudes. However, our research also revealed that human-AI collaboration could serve as an effective compromise. This collaborative approach, common in creative fields (Messer, 2024), significantly reduced the negative biases typically associated with purely AI-generated content.

These findings demonstrate that algorithm aversion in creative domains reflects deeper beliefs about creativity and effort, rather than simple skepticism toward AI capabilities. Such beliefs may be rooted in fundamental perceived differences between human and AI cognition (Monteith et al., 2024). Importantly, our identification of the machine heuristic as a significant moderator reveals that algorithm aversion in creative contexts varies systematically based on individual predispositions toward machine capabilities, rather than being uniform across users. This highlights the critical role of individual psychological factors in shaping human behavior with AI systems.

Our findings align with and extend research examining the gap between stated and actual authorship preferences. Like Raffloer and Green (Raffloer & Green, 2025), we find consistent bias against AI-labeled content when the source is disclosed. However, this bias contrasts with blind evaluation studies, where AI-written stories often receive more positive ratings than human-authored ones (Raffloer & Green, 2025). This paradox extends to real-world contexts: Zhou and Lee (Zhou & Lee, 2024) found that artists using generative AI on a large art platform were rated more favorably than their peers. Our research explains the psychological mechanism underlying this stated authorship bias. We demonstrate that negative reactions to AI labels operate through a specific pathway: AI attribution reduces perceived effort, which in turn diminishes perceived creativity and overall attitudes. This mechanism

helps reconcile seemingly contradictory findings in the literature. While people may actually prefer AI-generated content in blind comparisons, they systematically penalize the same content when it carries an AI label.

4.1. Theoretical implications

Our research makes three key contributions to algorithm aversion literature and human-AI interaction in creative contexts.

First, we advance the theoretical understanding of algorithm aversion by demonstrating how varying degrees of perceived human involvement, manipulated through authorship labels, influence individuals' perceptions of creative content. Our study design, which meticulously kept the underlying content identical across conditions, isolates the psychological impact of source attribution. Prior work has highlighted the importance of human involvement in mitigating algorithm aversion generally (Zhang et al., 2024). Our findings offer new insights into how individuals specifically evaluate different configurations of human and AI contributions when the output is creative. Individuals consistently prefer human-authored content, even over human-AI collaboration. This suggests that in creative domains, the mere presence of AI can trigger aversion. This aversion may stem from beliefs about fundamental differences in cognitive processes (Monteith et al., 2024) or a perceived lack of uniquely human qualities like authenticity (Messer, 2024) or humanness (Jia et al., 2024) in AI systems. Our findings contribute to a more nuanced theory of source attribution, helping to disentangle the effect of an authorship label from the effect of the actual content, which have been shown to operate in opposition to one another (Raffloer & Green, 2025).

Second, we identify a robust serial mediation pathway in creative contexts: perceived effort influences perceived creativity, which then shapes attitudes. This extends the work of Magni and colleagues. (Magni et al., 2023), who observed related effort-creativity links primarily in binary human vs. AI comparisons, by demonstrating these mechanisms at play within collaborative scenarios. This finding is crucial for theories of human-AI interaction as it suggests that for AI assistance to be positively received in creative endeavors, the collaborative process must still allow for the perception of significant human effort and creative input. The link between the viewers' lower perceived effort for AI-generated output and the actual creator's experience of reduced effort when using AI tools (Mei et al., 2025) further underscores the challenge in managing these effort perceptions. This effort-creativity pathway complements other psychological mechanisms identified in the literature, such as the 'meta-dehumanization' proposed by Chen and colleagues (Chen et al., 2024). It is plausible that these two mechanisms are related, as a layperson may perceive a purely calculative process as being both inherently 'effortless' and simultaneously dehumanizing.

Third, our primary theoretical contribution is the demonstration of the machine heuristic as a significant moderator of individual evaluations of AI-involved creative work. While other studies have identified general moderators in algorithm aversion, such as 'perceived AI capabilities' (Chen et al., 2025), or 'technology optimism' (Yang et al., 2025) our implementation of MH is more specific. It captures beliefs about AI capacity for inherently human-like qualities such as effort and creativity, rather than general competence alone. Previous research has explored MH in relation to trust in AI systems based on perceived accuracy or objectivity (Lucas & Nordgren, 2022; Sundar & Kim, 2019), our findings reveal a new dimension of this concept. We found that beliefs about an AI's inherent capabilities for effortful and creative tasks critically moderate the relationship between attributed authorship and perceived effort. Individuals high in MH attribute more effort to AI-generated output, challenging the common assumption of universally lower effort attributions to AI (Nie & Huang, 2023) and providing a key boundary condition for algorithm aversion theory in creative domains. This highlights the importance of considering individual cognitive heuristics when studying human behavior with AI systems, as these

pre-existing beliefs fundamentally shape how humans interpret and react to AI agents and their outputs.

4.2. Managerial implications

The findings of this research carry significant implications for practitioners, particularly managers and marketers in creative industries, navigating the integration of Artificial Intelligence and shaping consumer perceptions of their offerings. Our findings suggest three key areas for strategic action:

First, our research demonstrates that perceived effort plays a crucial mediating role in how individuals evaluate creative works attributed to different sources. When individuals attribute higher effort to the creation process, they perceive the output as more creative and develop more favorable attitudes. This finding held true across different attributed authorship types, suggesting that organizations should focus on transparently communicating the substantial work, ingenuity, and dedication invested in their creative processes, regardless of the degree of AI involvement. For instance, highlighting the iterative development, the complexity of human creative decisions, and the refinement stages involved in human-AI collaboration could enhance perceived effort and, consequently, creativity perceptions.

Second, organizations should strategically communicate the nature of their content creation process when AI systems are involved. The evidence suggests that explicitly highlighting human involvement in AI-generated or co-created content can enhance consumer perceptions of creativity and foster more favorable attitudes. Marketing campaigns and product descriptions should emphasize how human expertise guides, curates, or collaborates with AI capabilities, reinforcing that human creativity remains central to the product offering. For instance, streaming platforms could highlight how AI assists human writers in developing show concepts while emphasizing continued human creative direction and storytelling control, thereby framing the human-AI interaction positively.

Finally, managers should recognize that responses to AI involvement are not uniform and can vary based on their machine heuristic beliefs. This necessitates tailored communication strategies reflecting an understanding of different psychological profiles regarding AI systems. For individuals with high MH beliefs, emphasizing the advanced capabilities of the AI might bolster acceptance. Conversely, for individuals with lower MH beliefs, highlighting human oversight and significant involvement in AI-assisted processes will be essential to mitigate concerns about perceived effort and creativity. This segmentation approach can help organizations more effectively position their AI-integrated offerings and manage human perceptions of these emerging creative technologies. For contexts where machine-like qualities are desirable, such as in systematic or analytical tasks, managers can strategically trigger the machine heuristic by using interface cues that emphasize the system's non-human nature (Sundar & Kim, 2019).

4.3. Practical implications

The insights from this research extend beyond managerial strategies to offer practical guidance for various stakeholders interacting with or developing creative AI systems.

For AI tool designers and developers, our findings underscore the importance of considering how their systems are perceived in terms of effort and creative contribution. While efficiency is a key design goal, tools that completely obscure the creative process or appear to require minimal human input may inadvertently trigger negative perceptions as identified in our study. Future design iterations could explore features that allow for, or even highlight, significant human engagement, control, and refinement. For instance, interfaces that visually represent the iterative steps taken by a human user in guiding the AI, or that provide options for users to easily showcase their specific contributions within a human-AI collaboration, could help mitigate the perception of effortless,

purely machine-driven output and foster a greater sense of co-creation and perceived value.

Our research offers creative professionals like writers, artists, designers, and others working in creative fields a nuanced perspective on integrating AI into their workflows. While human-AI collaboration can partially mitigate negative perceptions compared to solely AI-authored work, the way this collaboration is framed and the degree of perceived human effort remain critical. Creators might consider transparently communicating not just that AI was used, but how it was used as a tool to augment their own skills and vision, emphasizing their continued creative direction and the effort involved in guiding, curating, and refining AI-generated elements. Understanding that viewers may hold varying machine heuristic beliefs also suggests that creators might need to adapt their disclosure strategies depending on their target audience, sometimes emphasizing the innovative AI capabilities, other times stressing their own mastery and control over the technology.

Finally, for viewers themselves, this research highlights the potential influence of authorship labels and underlying cognitive heuristics when evaluating AI-generated or co-created content. Recognizing that the attribution of "AI authorship" can shape perceptions of effort and creativity, potentially overshadowing an objective assessment of the content's merits, is crucial for developing critical media literacy. Rather than aiming to eliminate all discernment based on source, individuals can benefit from reflecting on their own machine heuristic beliefs about AI capabilities and limitations, along with their personal conceptions of what constitutes valuable creative work. This self-awareness can enable viewers to engage with content more consciously, acknowledging the role of the creator (whether human, AI, or a human-AI collaboration) while still critically evaluating the output itself, leading to more informed interactions with AI in the evolving creative landscape.

4.4. Limitations and future research

Our research has limitations that warrant consideration and provide opportunities for future investigation. First, methodologically, our experimental designs employed fictitious show descriptions in controlled settings. Specifically, by using identical content across authorship conditions and manipulating only the source label, we aimed to isolate the psychological effect of authorship labeling. While this approach enhances internal validity by controlling for content variations, it may not fully capture the complexity of real-world experiences where content genuinely created by AI versus humans would inherently differ. Study 1's within-subjects design, while helpful for controlling individual differences, may have created some demand effects by making the authorship manipulation more salient to participants.

Study 2's between-subjects design helped address the limitations of Study 1, but it introduced its own limitations. Specifically, we did not find a statistically significant direct effect between the human-AI collaboration and AI-only authorship type labels on perceived creativity or attitudes in Study 2, a result that contrasts with Study 1's findings. This discrepancy likely stems from the inherent differences between the designs. Study 1's within-subjects design allowed participants to directly compare different authorship types across multiple stimuli, making the distinctions more salient and reducing the influence of stimulus-specific noise. In contrast, Study 2's between-subjects design, with a single stimulus per participant, may have been less sensitive to these nuanced differences, even though it provided a cleaner test of the underlying mediating mechanisms. The mediation analysis in Study 2 further suggests that the lack of a direct effect might be due to opposing indirect effects: a negative indirect effect through creativity alone, and a positive indirect effect through the serial mediation of effort and creativity. While this statistical pattern helps explain the absence of a direct effect, it reinforces the idea that the within-subjects design was likely better suited for detecting subtle differences between authorship types in this specific context.

Regarding statistical power, a post-hoc analysis for Study 1

confirmed excellent power (>0.999) for the main effect of attributed authorship on attitudes. For Study 2, an a priori power analysis targeted a small-to-medium effect size (Cohen's $f = 0.15$), chosen due to the limited directly comparable effect size estimates for our novel stimuli and hypotheses. Ideally, a priori power calculations would have been conducted for both studies based on precise prior effect size estimates. While Study 2 was thus well-powered for its target, future research aiming to detect potentially smaller or more nuanced effects, particularly within moderated subgroups, might benefit from even larger samples or emerging literature to inform more precise a priori estimations.

Beyond these methodological limitations, our conceptual framework focused primarily on perceived effort and creativity as mediating mechanisms. However, other potential mediators, such as perceived authenticity, emotional connection, or perceived uniqueness, might also play important roles in how individuals evaluate AI-involved creative works. Furthermore, while we identified machine heuristic as an important moderator, other individual differences or contextual factors might influence these relationships.

Building on this need for real-world validation, future research could explore additional theoretical mechanisms underlying viewer responses to AI creativity. For instance, investigating how perceived authenticity or emotional resonance influence evaluations of AI-created content could enrich our understanding of algorithm aversion in creative contexts. The role of machine heuristic also deserves further investigation, particularly in understanding how these beliefs develop and whether they can be influenced by different types of AI experiences.

Finally, while our studies focused on U.S.-based participants, cultural attitudes toward AI and technology can vary significantly across different societies. Future research should explore these phenomena across diverse cultural settings to determine the generalizability of our findings and identify potential cultural moderators of algorithm aversion in creative contexts.

CRediT authorship contribution statement

Sigurd Birk Heimstad: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Anders Hauge Wien:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Tarje Gaustad:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Claude 3.7 (Anthropic) in order to improve writing, readability, and spelling. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration on ethical approval and informed consent

This study was exempt from formal ethics approval, according to the guidelines at UiT - The Arctic University of Norway. This study only collected age and gender details, and did not include names, contact information, geographic data, occupation, IP addresses, or any other variables that could reasonably identify a participant. All participants received an information page describing the purpose of the study, the voluntary nature of participation, anonymity and data handling. Participants provided their informed consent before proceeding with the survey.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.chbah.2025.100190>.

Data availability

Experiment1and2 (Original data) (OSF Open Science Framework)

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